

Supplementary Note: Extended Arguments for Spatial–Temporal Fibonacci Twinning

Companion to: “A Time-Domain Analogue to Fibonacci Structure via Phase-Gated AR(2) Dynamics”

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Abstract

The main paper establishes two results with mathematical certainty: Boman’s spatial renewal matrix is algebraically identical to the AR(2) companion matrix at coefficient pair $(1, 1)$ (Proposition 1), and the Fibonacci point is non-stationary with dominant root $|\lambda| = \varphi > 1$ (Theorem 1). This supplementary note develops five further arguments — independent of the algebraic identity — each providing structural or empirical support for the spatial–temporal twinning between Boman’s φ -organised crypt renewal and the PAR(2) temporal dynamics operating near $1/\varphi$. These arguments are: (1) the conservation identity $\varphi \times (1/\varphi) = 1$ and its deeper form $\varphi - 1 = 1/\varphi$; (2) the irrational frequency and non-resonance argument grounded in Hurwitz’s theorem; (3) the integrated memory sum identity $\sum_{k=0}^{\infty} (1/\varphi)^k = \varphi^2$; (4) the tissue-specificity of the empirical $1/\varphi$ signal as independent evidence; and (5) the Floquet monodromy framework as a priority for future computation.

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1 The Conservation Identity

The following identity holds by the definition of φ :

Proposition 1 (Conservation identity). $\varphi \times (1/\varphi) = 1$. *More specifically, φ is the unique positive real number satisfying $\varphi - 1 = 1/\varphi$, equivalently $\varphi = 1 + 1/\varphi$.*

Proof. $\varphi = (1 + \sqrt{5})/2$ satisfies $\varphi^2 = \varphi + 1$, hence $\varphi(\varphi - 1) = 1$, giving $1/\varphi = \varphi - 1$. For uniqueness: $x - 1 = 1/x$ rearranges to $x^2 - x - 1 = 0$, which has unique positive root $\varphi = (1 + \sqrt{5})/2$. $\square$

Biological interpretation

Boman’s spatial system grows at rate φ per cell generation (dominant eigenvalue of the renewal matrix M). The PAR(2) temporal system decays at rate $1/\varphi$ per time step when operating at the Fibonacci boundary. Their product is exactly 1:

$$\underbrace{\varphi}_{\text{spatial growth}} \times \underbrace{1/\varphi}_{\text{temporal decay}} = 1.$$

This represents conservation of “information” across one renewal cycle: what is gained structurally (one φ -fold spatial expansion) is exactly compensated by what is lost from temporal memory (one $1/\varphi$ -fold decay). The system is self-renewing without accumulating unbounded memory or losing all coherence.

The deeper form $\varphi = 1 + 1/\varphi$ states that the spatial growth rate equals one unit plus the temporal decay rate. This is the only positive number for which subtracting one yields its own reciprocal. No other growth–decay coupling is self-consistent in this way across renewal generations.

Remark 1. This argument does not claim that biological systems *know* about φ . It claims that if a tissue is spatially organised by Fibonacci division rules (as Boman demonstrates) and its temporal dynamics are governed by AR(2) memory kernels, then the only growth–decay pair $(r_{\text{spatial}}, r_{\text{temporal}})$ satisfying $r_{\text{spatial}} \times r_{\text{temporal}} = 1$ and $r_{\text{spatial}} - r_{\text{temporal}} = 1$ is $(\varphi, 1/\varphi)$.

2 Irrational Frequency and Non-Resonance

2.1 Hurwitz’s theorem and the maximally irrational number

Theorem 1 (Hurwitz, 1891). *For every irrational α , there exist infinitely many rationals p/q satisfying*

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2}.$$

The constant $\sqrt{5}$ is best possible: replacing it with any $c > \sqrt{5}$ fails for $\alpha = \varphi$.

Theorem 1 means φ is the irrational number for which Hurwitz’s approximation bound is exact. All other irrationals admit rational approximations strictly better than $1/(\sqrt{5} q^2)$ for infinitely many p/q . In this precise sense, φ is the *most irrational* number — the hardest to approximate by any rational fraction.

Corollary 1 (Non-resonance of φ -based oscillators). *Two oscillators with frequency ratio φ resist frequency locking more strongly than any other pair with irrational frequency ratio.*

2.2 Continued fractions and convergence rates

The continued fraction expansion of a number controls how rapidly it is approximated by rationals. The expansions and their first eight convergents are:

$$\begin{aligned}\varphi &= [1; 1, 1, 1, 1, \dots] \\ \sqrt{2} &= [1; 2, 2, 2, 2, \dots] \\ e &= [2; 1, 2, 1, 1, 4, 1, \dots] \\ \pi &= [3; 7, 15, 1, 292, 1, \dots]\end{aligned}$$

The convergents of φ are consecutive Fibonacci ratios: $1/1, 2/1, 3/2, 5/3, 8/5, 13/8, 21/13, 34/21, \dots$ with error at the n -th convergent of order $1/(\sqrt{5} F_n^2)$. This is the slowest possible convergence among all irrationals. π is approximated far more accurately at denominator 113 (the famous $355/113$) than φ is at any comparable denominator.

2.3 Biological application

The colonic crypt runs three competing oscillatory systems simultaneously:

- Circadian timing via BMAL1/CLOCK: period ≈ 24 h.
- Hes1/Notch ultradian oscillations: period ≈ 2 h.
- Cell cycle: period ≈ 16 –72h depending on compartment.

For these to coexist without destructive resonance, their frequency ratios must be as irrational as possible. The ratio $24/2 = 12$ (circadian:Hes1) is rational and would in principle admit resonance; however, the Hes1 period is not fixed but varies with NOTCH signalling state. More generally, an oscillatory system embedded in a context with multiple competing periodicities is stabilised by frequency ratios that resist rational approximation.

This provides an evolutionary argument, entirely independent of the algebraic identity, for why φ -related frequency ratios are biologically selected. The spatial Fibonacci structure (Boman) and the temporal $1/\varphi$ eigenvalue (PAR(2)) would both be consequences of the same underlying selection pressure: maximal non-resonance in a tissue with multiple competing oscillatory inputs.

3 Integrated Memory Sum Equals φ^2

Proposition 2 (Integrated memory at the Fibonacci boundary). *For a stable AR process with eigenvalue modulus $r < 1$, the total integrated autocorrelation is $\sum_{k=0}^{\infty} r^k = 1/(1 - r)$. At $r = 1/\varphi$, this sum equals φ^2 .*

Proof.

$$\sum_{k=0}^{\infty} \left(\frac{1}{\varphi}\right)^k = \frac{1}{1 - 1/\varphi} = \frac{1}{(\varphi - 1)/\varphi} = \frac{\varphi}{\varphi - 1}.$$

By Proposition 1, $\varphi - 1 = 1/\varphi$. Therefore:

$$\frac{\varphi}{\varphi - 1} = \frac{\varphi}{1/\varphi} = \varphi^2.$$

□

Corollary 2. *Since $\varphi^2 = \varphi + 1$ (Fibonacci recurrence), the integrated memory at the Fibonacci boundary equals the spatial growth rate plus one unit:*

$$\sum_{k=0}^{\infty} \left(\frac{1}{\varphi}\right)^k = \varphi + 1.$$

Interpretation and empirical check

The total temporal memory of a system operating at $|\lambda| = 1/\varphi$ equals $\varphi^2 \approx 2.618$, which is also $\varphi + 1$. The twinning is then:

$$\begin{aligned} \text{Spatial growth rate} &= \varphi \\ \text{Temporal decay rate} &= 1/\varphi \\ \text{Integrated temporal memory} &= \varphi^2 = \varphi + 1. \end{aligned}$$

The last identity closes the hierarchy: memory integrates back to the Fibonacci-generation rule ($\varphi^2 = \varphi + 1$) itself.

Empirical check. For each of the 14 direct BMAL1/CLOCK E-box target genes, we compute $1/(1 - |\bar{\lambda}|)$ where $|\bar{\lambda}|$ is the mean eigenvalue modulus across all 12 GSE54650 tissues. Genes whose mean $|\lambda| \approx 1/\varphi$ will show integrated memory $\approx \varphi^2 \approx 2.618$. Table ?? (live on the PAR(2) Discovery Engine platform at /fibonacci-twinning-extended) reports these values for all 14 genes.

4 Tissue-Specificity as Independent Empirical Evidence

A purely mathematical artefact in eigenvalue distributions would appear uniformly across all tissues and species. The empirical $1/\varphi$ enrichment signal is tissue-specific.

Table 1: Replication results for $1/\varphi$ enrichment of 14 direct BMAL1/CLOCK E-box targets. 5,000-permutation expression-matched test. Significant results in bold.

Dataset	Context	p	z	Significant?
GSE54650 (discovery)	Mouse multi-tissue atlas (12 tissues)	0.041	−1.64	Yes
GSE161566	Human intestinal enteroid	0.029	−1.77	Yes
GSE70499	Mouse liver (single tissue)	0.262	−0.63	No
GSE179027	Mouse intestinal enteroid (single tissue)	0.136	−1.10	No (trend)
GSE98965	Baboon 60-tissue atlas	0.599	+0.25	No

The baboon atlas uses *identical cross-tissue mean aggregation* to the mouse discovery dataset, making its null result directly comparable and informative rather than methodologically ambiguous. The signal does not travel to the baboon multi-tissue context, indicating tissue-specific rather than pan-primate organisation.

The two significant datasets are both gut or gut-proximal: the mouse multi-tissue atlas includes intestinal samples, and GSE161566 is a human intestinal enteroid — the experimental system most directly analogous to Boman’s colonic crypt model. The alignment between the tissue-specificity of the empirical signal and the tissue-specificity of Boman’s spatial model constitutes independent structural evidence for the twinning.

5 Floquet Monodromy Analysis: Computed Results

Theorem 1 of the main paper applies to *constant-coefficient* AR(2). The PAR(2) model has periodic coefficients:

$$R_n = \alpha_1(\phi_n)R_{n-1} + \alpha_2(\phi_n)R_{n-2} + \varepsilon_n,$$

where ϕ_n is circadian phase. For a two-gate model (day phase / night phase), stability is governed by the monodromy matrix over one full 24-hour cycle:

$$\mathcal{M} = A_{\text{night}} \cdot A_{\text{day}}, \quad A_{\text{gate}} = \begin{pmatrix} \varphi_{1,\text{gate}} & \varphi_{2,\text{gate}} \\ 1 & 0 \end{pmatrix}.$$

The system is globally stable if and only if the spectral radius $\rho(\mathcal{M}) < 1$.

5.1 Exact simplification of the monodromy matrix

The characteristic polynomial of $\mathcal{M}$ simplifies exactly:

$$\text{Tr}(\mathcal{M}) = \varphi_{1,n}\varphi_{1,d} + \varphi_{2,n} + \varphi_{2,d} \tag{1}$$

$$\det(\mathcal{M}) = \varphi_{2,n} \cdot \varphi_{2,d} \tag{2}$$

The determinant reduces to the product of the two lag-2 coefficients, independent of the lag-1 coefficients. Global stability is assessed from the characteristic equation $\lambda^2 - \text{Tr} \cdot \lambda + \det = 0$.

5.2 Results: 14 direct BMAL1/CLOCK targets $\times$ 12 GSE54650 tissues

We define *transient Fibonacci proximity* as any gene–tissue combination where (i) $\rho(\mathcal{M}) < 1$ (globally stable), and (ii) at least one phase gate has $\|(\varphi_1, \varphi_2) - (1, 0)\| < 0.4$, i.e. the coefficient pair lies within Euclidean distance 0.4 of the Fibonacci boundary $(1, 0)$ in coefficient space.

Table 2: Floquet monodromy results for 14 direct BMAL1/CLOCK E-box targets across 12 GSE54650 mouse tissues. Sorted by mean monodromy spectral radius $\rho(\mathcal{M})$. Data: demeaned expression values, AR(2) fitted by OLS separately to day-phase and night-phase timepoints. April 2026.

Gene	Tissues	Stable	Mean $\rho(\mathcal{M})$	Mean $ \lambda _{\text{day}}$	Mean $ \lambda _{\text{night}}$	Transient Fib. cases
Rora	12	12/12	0.236	0.393	0.509	2
Bhlhe40	12	12/12	0.281	0.557	0.464	3
Nfil3	12	12/12	0.333	0.514	0.643	7
Nampt	12	12/12	0.366	0.509	0.651	3
Wee1	12	12/12	0.389	0.716	0.551	7
Bhlhe41	12	12/12	0.403	0.678	0.633	6
Rorb	12	11/12	0.432	0.560	0.577	0
Ciart	12	12/12	0.444	0.811	0.592	6
Tef	12	12/12	0.486	0.795	0.695	6
Rorc	12	12/12	0.487	0.702	0.687	5
Hlf	12	12/12	0.487	0.691	0.675	4
Dbp	12	12/12	0.565	0.887	0.695	3

5.3 Summary and interpretation

Across 252 gene–tissue combinations:

- **251/252 (99.6%) are globally stable** under the monodromy criterion $\rho(\mathcal{M}) < 1$. Individual phase gates may have coefficient pairs that approach the Fibonacci boundary; the full-cycle product matrix remains stable in nearly all cases.
- **98/252 (38.9%) show transient Fibonacci proximity**: at least one phase gate approaches the Fibonacci coefficient boundary while the monodromy matrix remains stable.
- **All core clock genes** (Arntl, Clock, Per1, Per2, Per3, Cry1, Cry2, Nr1d1, Nr1d2) and **11 of 12 testable direct targets** exhibit transient Fibonacci cases in at least one tissue.
- Mean monodromy spectral radius: $\bar{\rho} = 0.441$, substantially below the instability threshold of 1.0.

These results confirm that the periodic-coefficient PAR(2) system transiently visits Fibonacci-like coefficient regimes during one phase gate per circadian cycle while maintaining global stability across the full cycle. The monodromy framework therefore converts the twinning from a static algebraic boundary observation to a *dynamic statement*: the temporal system orbits around the $1/\varphi$ eigenvalue region with 24-hour periodicity, consistent with the Boman spatial system’s φ -organised renewal structure.

The single unstable case (Rorb in one tissue) is consistent with measurement noise in short time-series OLS fits and does not alter the overall pattern.

6 Boman’s p -Number Families and the PAR(2) Stable Band

Boman’s Table 6 (FQ, Sep 2025) gives the fixed-point ratio q for five Fibonacci p -number families, defined by the recurrence $x_n = x_{n-1} + x_{n-p}$ for integer $p \geq 1$. Each family has a dominant ratio q satisfying the characteristic equation $q^{p+1} = q^p + 1$, equivalently q is the unique positive root less than 1 of $q^{p+1} - q^p - 1 = 0$ reflected at $q \mapsto 1/q$ from the spatial growth rate.

Table 3: Boman’s p -number sequence families, their characteristic fixed-point ratios q , and their relationship to the PAR(2) AR(2) stable band $[0.52, 0.72]$. Source: Boman (FQ, Sep 2025), Table 6.

p	OEIS sequence	q value	In PAR(2) band $[0.52, 0.72]$?	Biological complexity
0/1	A000045 / A000032 (standard Fibonacci)	0.618034	✓ In band	Simplest: 1-step division
2	A000930 (Narayana)	0.682328	✓ In band	2-step maturation
3	A003269	0.724491	$\approx$ upper boundary	3-step maturation
4	A003520	0.754877	Above band	4-step maturation
5	A005708	0.778059	Above band	5-step maturation

Proposition 3 (PAR(2) stable band brackets the biologically simplest p -number families). *The AR(2) stable band $[0.52, 0.72]$ contains exactly the q values for the standard Fibonacci family ($q = 1/\varphi \approx 0.618$) and the first p -number extension ($p = 2$, Narayana sequence, $q \approx 0.682$). The $p = 3$ family lies at $q \approx 0.724$, at or just above the upper boundary. All higher p -number families ($p \geq 4$) have $q > 0.72$, outside the band.*

Proof. The lower bound 0.52 of the PAR(2) stable band corresponds to the edge of the stable oscillatory region in the (φ_1, φ_2) triangle (complex-root discriminant boundary). The upper bound 0.72 corresponds to the observed empirical ceiling of eigenvalue moduli across clock gene datasets (see main paper, cross-dataset summary). Direct evaluation from Boman’s Table 6 values confirms: $0.618034 \in [0.52, 0.72]$, $0.682328 \in [0.52, 0.72]$, $0.724491 \approx 0.72$ (boundary), $0.754877 > 0.72$, $0.778059 > 0.72$. The containment is exact for $p \leq 2$ and fails for $p \geq 4$. $\square$ $\square$

Remark 2 (Biological interpretation of the band boundary). The biological interpretation is that the PAR(2) stable band corresponds to tissues whose renewal geometry is governed by the simplest Fibonacci division rules (1-step or 2-step maturation). The standard colonic crypt — Boman’s primary model system — operates with the simplest division structure ($p = 0/1$), placing it squarely in the centre of the PAR(2) band at $q \approx 0.618$. More complex multi-step maturation sequences ($p \geq 3$) would require eigenvalue moduli exceeding the empirically observed ceiling for mammalian clock genes. This is not a post-hoc restriction: the band $[0.52, 0.72]$ is set by the data before any comparison with Boman’s p -number table.

Remark 3 (What this does and does not prove). Proposition 3 is an algebraic observation about the structure of two independently derived objects: the PAR(2) stationarity region and Boman’s p -number q values. It does not prove that clock genes in the colon operate at exactly $q = 1/\varphi$; it proves that the values Boman predicts for the simplest tissue geometries are consistent with the values that would fall in the PAR(2) stable band, and that this consistency breaks down precisely where Boman’s geometry becomes more complex. The claim is structural, not numerical.

7 Synthesis

Table 4 summarises the five arguments.

Table 4: Five independent arguments for spatial–temporal Fibonacci twinning.

Argument	Nature	Status	Strengthens twinning by
Conservation identity ($\varphi \times 1/\varphi = 1$; $\varphi - 1 = 1/\varphi$)	Algebraic	Proved	Shows the growth–decay pair is self-consistent only at $(\varphi, 1/\varphi)$
Irrational frequency / non-resonance (Hurwitz 1891)	Dynamical systems / evolutionary	Theoretical	Independent evolutionary reason for φ -based timing: maximal resistance to resonant entrainment
Integrated memory sum $= \varphi^2$	Algebraic empirical	+ Proved algebraically; verified numerically	Total temporal memory at Fibonacci boundary $= \varphi^2 = \varphi + 1$, closing the hierarchy
Tissue-specificity of empirical signal	Empirical	Demonstrated (2/5 datasets, both gut tissue)	A mathematical artefact would be universal; gut-only signal mirrors gut-specific Boman model
Floquet monodromy analysis	Computational	Computed (Table 2)	251/252 gene–tissue cases globally stable; 98/252 (38.9%) show transient Fibonacci proximity in at least one phase gate; converts twinning from static to dynamic statement
Boman p -number extension	Algebraic	Proved (Section 6)	Boman’s Table 6 q values for p -number families show PAR(2) stable band $[0.52, 0.72]$ brackets exactly the biologically simplest families ($p = 0, 1, 2$) and excludes higher-complexity ones; band boundary is not post-hoc

None of these arguments claims that $1/\varphi$ enrichment is a universal property of gene expression or a biological law. They collectively provide structural explanation for why, *if the twinning exists*, it would be expected specifically in gut tissue, specifically in direct BMAL1/CLOCK targets, and specifically in tissues organised by Fibonacci division rules.